

| 19    | B                           | <p>Table shows <math>\sin \theta = \frac{n\lambda}{d}</math></p> <table border="1" data-bbox="524 222 1830 743"> <thead> <tr> <th>order</th><th>violet (400 nm)</th><th>red (700 nm)</th></tr> </thead> <tbody> <tr> <td>1</td><td><math>\frac{400 \text{ nm}}{d}</math></td><td><math>\frac{700 \text{ nm}}{d}</math></td></tr> <tr> <td>2</td><td><math>\frac{800 \text{ nm}}{d}</math></td><td><math>\frac{1400 \text{ nm}}{d}</math></td></tr> <tr> <td>3</td><td><math>\frac{1200 \text{ nm}}{d}</math></td><td><math>\frac{2100 \text{ nm}}{d}</math></td></tr> </tbody> </table> <p>Since <math>\frac{1200 \text{ nm}}{d} &lt; \frac{1400 \text{ nm}}{d}</math>, 3<sup>rd</sup> order spectrum overlaps with the 2<sup>nd</sup> order spectrum.</p> | order | violet (400 nm) | red (700 nm) | 1 | $\frac{400 \text{ nm}}{d}$ | $\frac{700 \text{ nm}}{d}$ | 2 | $\frac{800 \text{ nm}}{d}$ | $\frac{1400 \text{ nm}}{d}$ | 3 | $\frac{1200 \text{ nm}}{d}$ | $\frac{2100 \text{ nm}}{d}$ |
|-------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------|--------------|---|----------------------------|----------------------------|---|----------------------------|-----------------------------|---|-----------------------------|-----------------------------|
| order | violet (400 nm)             | red (700 nm)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |       |                 |              |   |                            |                            |   |                            |                             |   |                             |                             |
| 1     | $\frac{400 \text{ nm}}{d}$  | $\frac{700 \text{ nm}}{d}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |                 |              |   |                            |                            |   |                            |                             |   |                             |                             |
| 2     | $\frac{800 \text{ nm}}{d}$  | $\frac{1400 \text{ nm}}{d}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |                 |              |   |                            |                            |   |                            |                             |   |                             |                             |
| 3     | $\frac{1200 \text{ nm}}{d}$ | $\frac{2100 \text{ nm}}{d}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |                 |              |   |                            |                            |   |                            |                             |   |                             |                             |
| 20    | B                           | The constant electric force on the electron is pointing vertically downwards, in the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |                 |              |   |                            |                            |   |                            |                             |   |                             |                             |